Chapter 12 · Differentiation

Gradient and first principles

Understand the derivative as a gradient function and derive simple derivatives from the limit definition.

Differentiation rules

Differentiate powers, sums and expressions rewritten as powers. Find second derivatives.

Using derivatives

Find tangents and normals, increasing and decreasing intervals, and stationary points. Sketch functions and their derivatives.

Modelling

Use rates of change and constraints to solve maximum and minimum problems.

Gradient Function

Luisa · 3

For a straight line, the gradient is constant:

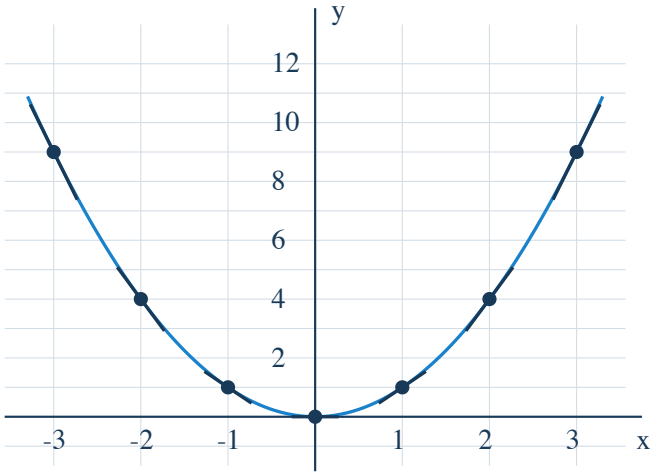
$y = 3x + 2$



For a curve, the gradient varies. We can no longer use a single constant for the gradient. We want an expression in x that gives the gradient at any value of x: the gradient function.

At GCSE, you found the gradient of a curve at a particular point by drawing a tangent.

$y = x^2$



x	-3	-2	-1	0	1	2	3
Gradient	1.	2.	3.	4.	5.	6.	7.

Use the relationship between x and the gradients to find the gradient function.

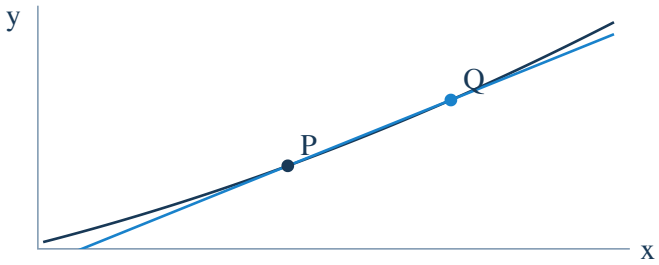
8.

Finding the Gradient Function

Luisa · 4

Is there a method to find the gradient function without drawing lots of tangents and hoping to spot the rule?

To approximate the gradient on $y = x^2$ when $x = 5$, choose a second point slightly to the right. First use $x = 6$, then find the gradient between P and Q.



Coordinates of P

1.

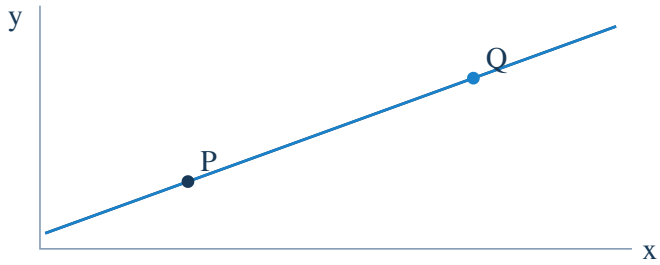
Coordinates of Q

2.

3.

As the second point gets closer, the gradient between the points becomes a better approximation of the true gradient. Now use $x = 5.01$.

Magnified view near $x = 5$



P remains $(5, 25)$. Find the coordinates of Q.

4.

5.

The actual gradient at $x = 5$ is 10, so the second calculation gives a much closer approximation.

Finding the Gradient Function **Luisa · 5**

For $y = x^2$, replace a numerical step by a small nonzero change h. Find the gradient at a general point.

Starting point

P

Nearby point

Q

Form the difference quotient

Expand and cancel, for h ≠ 0

Take the limit

lim means the limiting value as the input approaches a value. We simplify before taking the limit; we never divide by zero.

Finding the Gradient Function **Luisa · 6**

The gradient function, or derivative

$$f\prime(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

This is differentiation from first principles. When the limit exists, it gives the gradient of the tangent at x.

Leibniz notation

$$y = x^2 \quad \Rightarrow \quad \frac{dy}{dx} = 2x$$

Lagrange notation

$$f(x) = x^2 \quad \Rightarrow \quad f\prime(x) = 2x$$

Changes and limits

A finite change is often written Δx or δx. Taking the limit of the difference quotient gives dy/dx.

$$\lim_{\delta x \to 0} \frac{f(x+\delta x) - f(x)}{\delta x}$$

Newton notation

A dot usually denotes differentiation with respect to time: $\dot{y} = dy/dt$. For $y = t^2$, $\dot{y} = 2t$.

Keep the variable of differentiation clear.

Example

Luisa · 7

A(4,16) lies on $y = x^2$, where the gradient is g.

(a) Show that $g = \lim_{h \to 0}(8 + h)$.

a

(b) Deduce the value of g.

b

Test Your Understanding

Luisa · 8

Prove from first principles that the derivative of x^4 is $4x^3$.

The relevant row of Pascal's triangle is 1, 4, 6, 4, 1.

Proof

After cancellation, every term containing a positive power of h tends to zero.

Exercise 12B – Pearson Pure Y1 page 261-262 **Luisa · 9**

5. Prove from first principles that the derivative of $6x$ is 6.

Use the definition

$$f\prime(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Proof

Exercise 12B – Pearson Pure Y1 page 261-262 **Luisa · 10**

6. Prove from first principles that the derivative of $4x^2$ is $8x$.

Use the definition

$$f\prime(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Proof

Exercise 12B – Pearson Pure Y1 page 261-262 **Luisa · 11**

7. Prove from first principles that the derivative of ax^2 is $2ax$. Here a is constant.

Use the definition

$$f\prime(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Proof

Exercise 12B – Pearson Pure Y1 page 261-262 **Luisa · 12**

Challenge: $f(x) = 1/x$, with $x \neq 0$.

(a) Use first principles to show that $f\prime(x) = \lim_{h \to 0} \frac{-1}{x^2 + xh}$.

a

(b) Deduce that $f\prime(x) = -1/x^2$.

b

Just for your interest... — Luisa · 13

Why can we not immediately set $h = 0$ in $\lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$?

If we substitute $h = 0$ straight away:
$\lim_{h \to 0} \frac{(x+0)^2 - x^2}{0} = \lim_{h \to 0} \frac{0}{0}$.
Wait: this does not determine the limit.

Thankfully, there are techniques that turn a limit expression into one without an indeterminate form.

In this example, expand and simplify:
$\frac{(x+h)^2 - x^2}{h} = 2x + h$, for $h \neq 0$.
Hence $\lim_{h \to 0}(2x + h) = 2x$; substituting here is valid. Other techniques are more advanced.

This is important when sketching harder functions.

Further source resource: DrFrostMaths, resource 163.

$\frac{0}{0}$ is an indeterminate form.

Some expressions have limits that grow without bound, such as $\frac{1}{h}$ as $h \to 0^+$. But an indeterminate form alone does not determine a limit.

Zero divided by a non-zero number is zero; dividing by a number approaching zero can produce unbounded values. These observations do not resolve $\frac{0}{0}$.

Can you guess other indeterminate forms: expressions whose limiting values cannot be determined from the form alone?

1.

Differentiating x^n — Luisa · 14

The power rule

$$\frac{d}{dx}(ax^n) = nax^{n-1}$$

Multiply by the power and reduce the power by one. Here a and n are constants; use a domain where the function is defined.

1. $y = x^5$

1

2. $f(x) = x^{1/2}$

2

3. $y = 2x^6$

3

4. $f(x) = x/x^4$

4

5. $y = \sqrt{x^6}$, for $x \geq 0$.

5

The rule does not apply to 2^x: its exponent is variable, not constant.

Test Your Understanding **Luisa · 15**

Differentiate each expression. In question 4, a is constant.

1. $y = x^7$

1

3. $f(x) = \frac{x^{1/2}}{x^2}$

3: rewrite

4. $y = ax^a$

4

2. $y = 3x^{10}$

2

3: derivative

5. $f(x) = \sqrt{49x^7}$

5: rewrite

5: derivative

Differentiating Multiple Terms **Luisa · 16**

Differentiate a sum or difference term by term.

$$\frac{d}{dx}[f(x) + g(x)] = f\prime(x) + g\prime(x)$$

Example: $y = x^2 + 4x + 3$.

$\frac{d}{dx}(x^2)$

$\frac{d}{dx}(4x)$

$\frac{d}{dx}(3)$

3

Combine the derivatives

Result

A linear term has a constant derivative. A constant term has derivative zero because its graph is horizontal.

Quickfire Questions — Luisa · 17

Differentiate each function. Here a and p are constants.

1. $y = 2x^2 - 3x$

1

2. $y = 4 - 9x^3$

2

3. $y = 5x + 1$

3

4. $y = ax$

4

5. $y = 6x - 3 + px^2$

5

Harder Example — Luisa · 18

Let $f(x) = 4x^2 - 8x + 3$.

(a) Find the gradient at $(\frac{1}{2}, 0)$.

a

(b) Find the point where the gradient is 8.

b

(c) Find the gradients where the curve meets $y = 4x - 5$.

c

A known x gives the gradient. A known gradient gives an equation for x; substitute into the original function to find y.

Test Your Understanding **Luisa · 19**

Let $f(x) = x^2 - 4x + 2$.

(a) Find the gradient at $(1, -1)$.

a

(b) Find the point where the gradient is 5.

b

(c) Find the gradients where the curve meets $y = 2 - x$.

c

A known x gives the gradient. A known gradient gives an equation for x; substitute into the original function to find y.

Exercise 12D – Pearson Pure Y1 page 265-266 **Luisa · 20**

5. Find the gradients of $y = x^2 - 5x + 10$ where the curve meets $y = 4$.

Find the intersections

Differentiate

Evaluate at each intersection

6. Find the gradients of $y = 2x^2$ where the curve meets $y = x + 3$.

Find the intersections

Differentiate

Evaluate at each intersection

7. $f(x) = x^2 - 2x - 8$.

a. Sketch $y = f(x)$.

b. On the same axes, sketch $y = f\prime(x)$.

c. Explain why the turning point of $y = f(x)$ has the same x-coordinate as the point where $y = f\prime(x)$ crosses the x-axis.

a.

b.

c.

Differentiating Harder Expressions

Rewrite as a sum of powers of x before differentiating.

1. Turn roots into powers

$y = \sqrt{x}$

1: rewrite

1: derivative

2. Turn roots into powers

$y = 1/\sqrt[3]{x}$

2: rewrite

2: derivative

3. Split fractions

$y = \dfrac{x^2 + 3}{\sqrt{x}}$

3: rewrite

3: derivative

4. Expand brackets

$y = x^2(x - 3)$

4: rewrite

4: derivative

5. Keep constant factors

$y = \dfrac{1}{3x}$

5: rewrite

5: derivative

Test Your Understanding

Differentiate each expression.

1. $y = \frac{1}{\sqrt{x}}$

1: rewrite

1: derivative

2. $y = \frac{2+x^3}{x^2}$

2: rewrite

2: derivative

3. $y = \frac{1+2x}{3x\sqrt{x}}$

3: rewrite

3: derivative

Exercise 12E – Pearson Pure Y1 page 267-268 **Luisa · 25**

6. $f(x) = \frac{12}{p\sqrt{x}} + x$, where p is a nonzero real constant and x > 0. Given $f\prime(2) = 3$, find p in the form $a\sqrt{2}$, where a is rational.

Rewrite and differentiate

Use the given gradient

Exercise 12E – Pearson Pure Y1 page 267-268 **Luisa · 26**

7. $f(x) = (2 - x)^9$.

(a) Find the first three terms in ascending powers of x.

a

(b) For small x, ignore terms in x^2 and higher in the derivative. Show that $f\prime(x) \approx 9216x - 2304$.

b

Extension — Luisa · 27

[MAT 2013 1E]

The expression $\frac{d^2}{dx^2}\left[(2x-1)^4(1-x)^5\right] + \frac{d}{dx}\left[(2x+1)^4(3x^2-2)^2\right]$

is a polynomial of degree:

9

8

7

less than 7

The second derivative means differentiate twice. It comes later in this chapter.

Finding equations of tangents — Luisa · 28

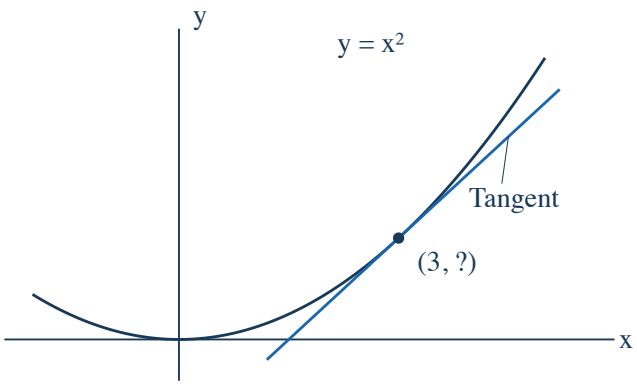
Find the equation of the tangent to the curve $y = x^2$ when $x = 3$.

We want to use $y - y_1 = m(x - x_1)$ for the tangent, because it is a straight line.

Therefore we need:

• A point (x_1, y_1).

• The gradient m.



Gradient function: 1.

Gradient when $x = 3$: 2.

y-value when $x = 3$: 3.

So the equation of the tangent is: 4.

Finding equations of normals

Luisa · 29

Find the equation of the normal to $y = x^2$ at x $= 3$.

The normal is perpendicular to the tangent at the point of contact.

Point and tangent gradient

Normal gradient

The tangent is y $- 9 = 6$(x $- 3$).

Equation of the normal

Check whether the question asks for a tangent or a normal. Their gradients are negative reciprocals when both are finite and nonzero.

Test Your Understanding

Luisa · 30

Find the normal to $y = x + 3\sqrt{x}$ at x $= 9$.

Point on the curve

Tangent and normal gradients

Normal equation

Exercise 12F – Pearson Pure Y1 page 269-270 **Luisa · 31**

3. Find where the tangent to $y = x^2 + 1$ at $(2,5)$ meets the normal to the curve at $(1,2)$.

Tangent at $(2,5)$

Normal at $(1,2)$

Solve simultaneously

3

Exercise 12F – Pearson Pure Y1 page 269-270 **Luisa · 32**

4. Find the normals to $y = x + x^3$ at $(0,0)$ and $(1,2)$, and find their point of intersection.

Normal at $(0,0)$

Normal at $(1,2)$

Intersection

5. For $f(x) = 12 - 4x + 2x^2$, find the tangent and normal at x $= -1$.

Point and gradient

Tangent

Normal

6. P has x-coordinate $\frac{1}{2}$ on $y = 2x^2$. The normal at P meets the curve again at Q. Find Q. (6 marks)

Normal at P

Second intersection

Extension

1. [STEP I 2005 Q2] On $y^2 = 4x$, let $P = (p^2, 2p)$ and $Q = (q^2, 2q)$, where p and q are nonzero and distinct.

(a) Show that their tangents meet at $R = (pq, p + q)$.

(b) Their normals meet at S. If (1,0) lies on PQ, show that $S = (p^2 + q^2 + 1, p + q)$ and that PSQR is a rectangle.

Worked solution on the next slide.

2. [STEP I 2012 Q4] On $xy = \frac{1}{2}$, let $P = (p, \frac{1}{2p})$, $Q = (q, \frac{1}{2q})$, with distinct positive p and q.

(a) Show that the tangents meet at $T = (\frac{2pq}{p+q}, \frac{1}{p+q})$.

(b) When $pq = \frac{1}{2}$, find the intersection N of the normals. Show that T and N lie on y = x and that their distances from O have constant product.

2(a)

2(b)

Useful identity

$p^3 - q^3 = (p - q)(p^2 + pq + q^2)$

Solutions to Extension Question 1

For $y^2 = 4x$, implicit differentiation gives $2y\,dy/dx = 4$, so $dy/dx = 2/y$. This works on either branch when y ≠ 0.

Tangents at P and Q

Use the chord condition

Normals at P and Q

Show that PSQR is a rectangle

Increasing and Decreasing Functions

Increasing and decreasing

A function is increasing on an interval when larger inputs give values that are at least as large. For a differentiable function, $f\prime(x) \geq 0$ throughout an interval ensures this.

Likewise, $f\prime(x) \leq 0$ throughout an interval ensures the function is decreasing.

Strictly increasing

If $f\prime(x) > 0$ throughout an interval, the function is strictly increasing. This is sufficient, not necessary: x^3 is strictly increasing although its derivative is zero at $x = 0$.

A curve has a local maximum at $(2,3)$ and a local minimum at $(4,-1)$. Its gradient is positive outside these turning points and negative between them. State its increasing and decreasing intervals.

Increasing: left

Decreasing

Increasing: right

Interval notation: $[2, 4] = \{x : 2 \leq x \leq 4\}$. Square brackets include the endpoints.

Examples

1. Show that $f(x) = x^3 + 6x^2 + 21x + 2$ is increasing for every real x.

1

2. Find the interval where $g(x) = x^3 + 3x^2 - 9x$ is decreasing.

2

Test Your Understanding — Luisa · 39

1. Show that $f(x) = x^3 + 16x - 2$ is increasing for every real x.

1

2. Find the interval where $g(x) = x^3 + 6x^2 - 135x$ is decreasing.

2

Exercise 12G – Pearson Pure Y1 page 271 — Luisa · 40

3. Show that $f(x) = 4 - x(2x^2 + 3)$ is decreasing for all real x.

Expand and differentiate

Determine the sign

4. $f(x) = x^2 + px$ is increasing on $[-1,1]$.

(a) Find one possible value of p.

a

(b) Is this the only possible value? Justify your answer.

b

Second Order Derivatives **Luisa · 42**

The second derivative is the derivative of the first derivative. Keep the variable of differentiation consistent.

Leibniz notation	Lagrange notation	Newton notation: time
$y = x^4$	$f(x) = x^4$	$y = t^4$
$\frac{dy}{dx} = 4x^3$	$f\prime(x) = 4x^3$	$\dot{y} = 4t^3$
$\frac{d^2y}{dx^2} = 12x^2$	$f\prime\prime(x) = 12x^2$	$\ddot{y} = 12t^2$

The second derivative describes how the gradient changes. We will use its sign to help classify stationary points.

Just for your interest... — Luisa · 43

Why is the second derivative written this way?

For $y = x^3 + 1$, the symbol d/dx is an operator: it means "differentiate with respect to x".

$$\frac{d}{dx}(x^3 + 1) = 3x^2$$

Apply the operator twice

$$\frac{d}{dx}\left(\frac{d}{dx}(y)\right) = \frac{d^2y}{dx^2}$$

Example

$$\frac{d}{dx}(3x^2) = 6x$$

$$\frac{d^2y}{dx^2} = 6x$$

The superscripts indicate two applications of differentiation with respect to x. Do not treat the notation as an ordinary fraction to be squared.

Examples — Luisa · 44

1. If $y = 3x^5 + \frac{4}{x^2}$, find d^2y/dx^2.

1: first derivative

1: second derivative

2. If $f(x) = 3\sqrt{x} + \frac{1}{2\sqrt{x}}$, find $f''(x)$.

2: first derivative

2: second derivative

Test Your Understanding — Luisa · 45

If $y = 5x^3 - \frac{x}{3\sqrt{x}}$, find d^2y/dx^2.

Rewrite as powers	Differentiate once	Differentiate again

Exercise 12H – Pearson Pure Y1 page 272 — Luisa · 46

3. Given $y = (2x - 3)^3$, find x when $d^2y/dx^2 = 0$.

3

3

Exercise 12H – Pearson Pure Y1 page 272 **Luisa · 47**

4. For $y = px^3 - 3px^2 + x^2 - 4$, the second derivative at x = 2 is -1. Find p.

4

4

Stationary/Turning Points **Luisa · 48**

A stationary point is a point where the derivative is zero. It may be a local maximum, a local minimum or a stationary point of inflection.

Find the stationary points of $y = x^3 + 6x^2 - 135x$.

Solve for the x-coordinates

Find the y-coordinates

Use the original function to find y, not the derivative. We will next classify the two points.

1. Find the least value of
$f(x) = x^2 - 4x + 9$.

1. Method 1

1. Method 2

Note: Method 2 applies to quadratic functions. For other functions, use differentiation.

2. Find the turning point of
$y = \sqrt{x} - x$.

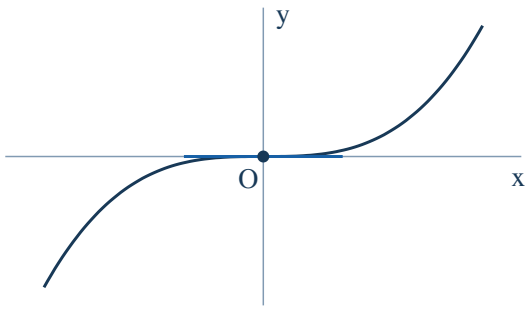
2.

Points of Inflection

A third type of stationary point is a stationary point of inflection.

A point of inflection is where a curve changes concavity: it curves one way before the point, then the other way after it.

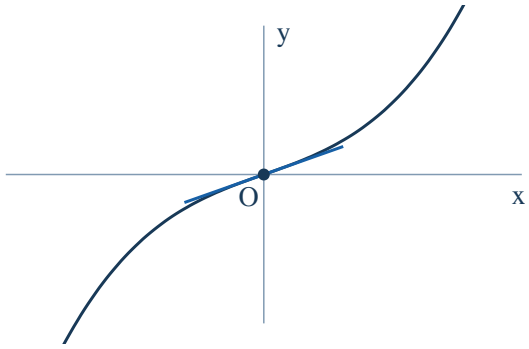
Stationary point of inflection



At the point, $f\prime(x) = 0$.
The tangent is horizontal; the concavity changes.

The terms convex and concave are also used in optics. Terminology varies; the key feature here is the change in concavity.

Non-stationary point of inflection

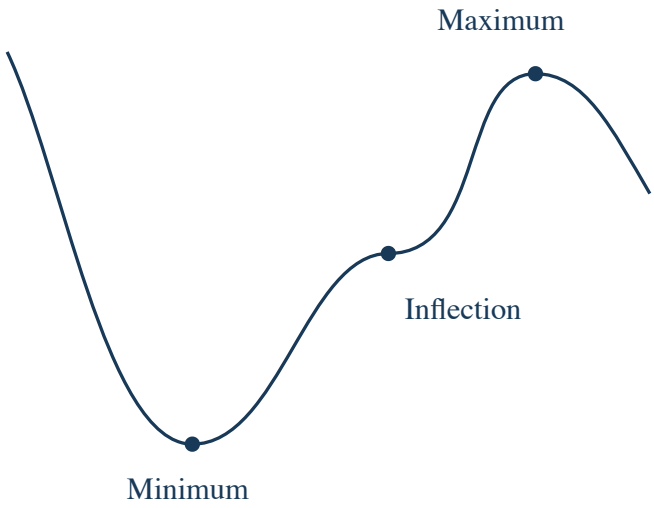


Here $f\prime(x) \neq 0$.
Not every point of inflection is stationary.

A stationary point of inflection is sometimes called a saddle point.

How do we tell what type of stationary point? **Luisa · 51**

Method 1: Compare the gradient just before and just after the stationary point.



At a minimum or maximum the gradient changes sign. At a stationary point of inflection it is zero at the point without a change of sign.

Local minimum - gradient

Before point	At point	After point
1.	2.	3.

Local maximum - gradient

Before point	At point	After point
4.	5.	6.

Stationary point of inflection - gradient

Before point	At point	After point
7.	8.	9.

How do we tell what type of stationary point? **Luisa · 52**

Find and classify the stationary point of $y = x^4 - 32x$ using the gradient on either side.

Locate the point

Just before: x = 1.9

At the point: x = 2

Just after: x = 2.1

Conclusion

Method 2: Using the second derivative

The second derivative is the rate of change of the gradient.

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

Near a local maximum

The gradient falls from positive to zero to negative. When the second derivative at the stationary point is negative, the point is a local maximum.

Near a local minimum

The gradient rises from negative to zero to positive. When the second derivative at the stationary point is positive, the point is a local minimum.

First locate a stationary point by solving $\mathrm{dy}/\mathrm{dx} = 0$. Then evaluate the second derivative at that x-coordinate.

Method 2: Using the second derivative

The second derivative test at a stationary point

$f''(x) > 0$

Local minimum: the gradient is increasing.

$f''(x) < 0$

Local maximum: the gradient is decreasing.

$f''(x) = 0$

Inconclusive. Check the gradient on either side or use another argument.

Example: $y = x^4 - 32x$ has a stationary point at $(2, -48)$.

Example

A zero second derivative does not automatically mean a point of inflection.

Test Your Understanding — Luisa · 55

The curve $y = x^2 - 32\sqrt{x} + 20$, $x > 0$, has a stationary point P. Use calculus:

a. Find the coordinates of P. (6 marks)

b. Determine the nature of P. (3 marks)

a.

b.

Sketching Graphs — Luisa · 56

In Chapter 4, we used intercepts and behaviour as $x \to \infty$ and $x \to -\infty$ to sketch graphs. Now we can also find stationary and turning points.

[Textbook] By first finding the stationary points, sketch $y = \frac{1}{x} + 27x^3$.

1.

2.

3.

4.

Exercise 12I – Pearson Pure Y1 page 276 **Luisa · 57**

5. $f(x) = x^3 - 3x^2 + 3x$.

(a) Find the stationary point.

a

(b) Determine its nature.

b

(c) Sketch the graph.

c

Sketch

Exercise 12I – Pearson Pure Y1 page 276 **Luisa · 58**

6. $f(x) = 27 - 2x^4$.

(a) Find and classify the stationary point.

a

(b) State the range of f.

b

Here $f''(0) = 0$, so the second derivative test alone is inconclusive. The gradient sign change establishes the maximum.

7. $f(x) = x^4 + 3x^3 - 5x^2 - 3x + 1$.

a. Find the coordinates of the stationary points and determine the nature of each.

b. Sketch $y = f(x)$.

a.

b.

Extension **Luisa · 60**

1. [MAT 2014 1C] The cubic
$y = kx^3 - (k+1)x^2 + (2-k)x - k$
has a turning point that is a minimum at $x = 1$ precisely for:
A) $k > 0$ B) $0 < k < 1$ C) $k > \frac{1}{2}$
D) $k < 3$ E) all values of k.

2. [MAT 2004 1B] Find the smallest value of
$f(x) = 2x^3 - 9x^2 + 12x + 3$
in the range $0 \le x \le 2$.

3. [MAT 2001 1E] The maximum gradient of
$y = x^4 - 4x^3 + 4x^2 + 2$
in $0 \le x \le 2\frac{1}{5}$ occurs when:
A) $x = 0$ B) $x = 1 - \frac{1}{\sqrt{3}}$
C) $x = 1 + \frac{1}{\sqrt{3}}$ D) $x = 2\frac{1}{5}$.

4. [STEP I 2007 Q8] A curve is given by
$y = ax^3 - 6ax^2 + (12a+12)x - (8a+16)$,
where a is a real number. Show that it touches $y = x^3$ at $(2, 8)$.
Determine the coordinates of any other point of intersection of the curves.
Sketch both curves on the same axes when:
(i) $a = 2$;
(ii) $a = 1$;
(iii) $a = -2$.

Hint: When two curves touch, their y-values must match. What else must also match?

Worked solutions follow on slides 61–63.

1. [MAT 2014 1C] The cubic
$y = kx^3 - (k+1)x^2 + (2-k)x - k$
has a turning point that is a minimum at $x = 1$ precisely for:
A) $k > 0$ B) $0 < k < 1$ C) $k > \frac{1}{2}$
D) $k < 3$ E) all values of k.

1.

2. [MAT 2004 1B] Find the smallest value of
$f(x) = 2x^3 - 9x^2 + 12x + 3$
in the range $0 \leq x \leq 2$.

2.

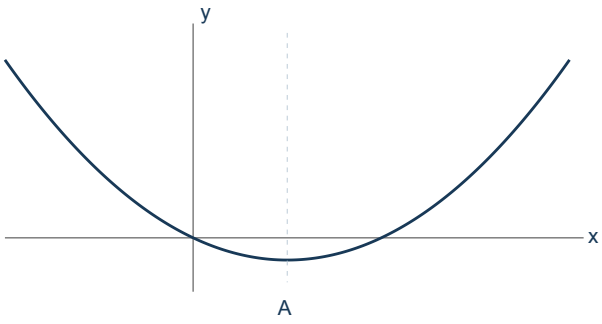
3. [MAT 2001 1E] The maximum gradient of $y = x^4 - 4x^3 + 4x^2 + 2$ in $0 \leq x \leq 2\frac{1}{5}$ occurs when:
A) $x = 0$ B) $x = 1 - \frac{1}{\sqrt{3}}$ C) $x = 1 + \frac{1}{\sqrt{3}}$ D) $x = 2\frac{1}{5}$.

3.

4. [STEP I 2007 Q8] A curve is given by:
$y = ax^3 - 6ax^2 + (12a + 12)\,x - (8a + 16)$
where a is a real number. Show that this curve touches the curve with equation $y = x^3$ at $(2, 8)$. Determine the coordinates of any other point of intersection of the two curves.
(i) Sketch on the same axes these two curves when $a = 2$.
(ii) ... when $a = 1$
(iii) ... when $a = -2$

4

Original curve: $y = f(x)$



Gradient function: $y = f\prime(x)$

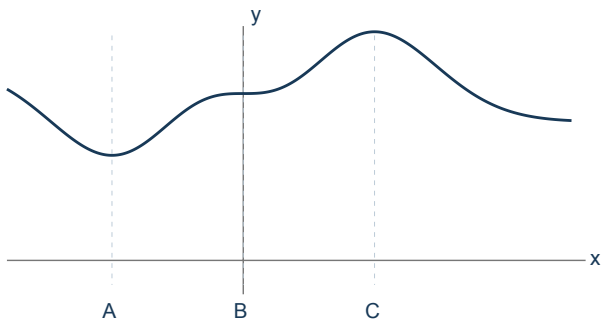
Gradient graph

The curve is a parabola with its minimum at A. Sketch its gradient function.

The derivative of a quadratic is a straight line. Its x-intercept is directly below the turning point.

A Harder One — Luisa · 65

Original curve: $y = f(x)$



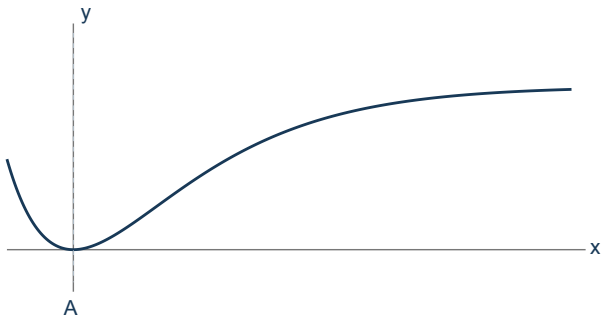
Gradient function: $y = f\prime(x)$

Gradient graph

Sketch the gradient function. A is a local minimum, B a stationary inflection and C a local maximum.

Test Your Understanding — Luisa · 66

Original curve: $y = f(x)$



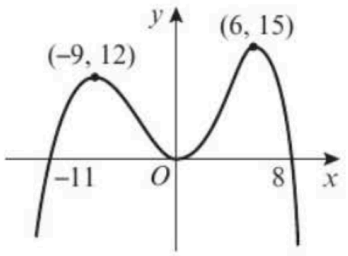
Gradient function: $y = f\prime(x)$

Gradient graph

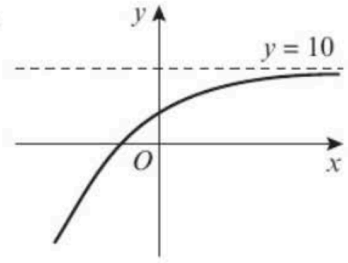
Sketch the gradient function of the curve shown. The curve has a minimum at A and approaches a horizontal asymptote to the right.

1 For each graph given, sketch the graph of the corresponding gradient function on a separate set of axes. Show the coordinates of any points where the curve cuts or meets the x-axis, and give the equations of any asymptotes.

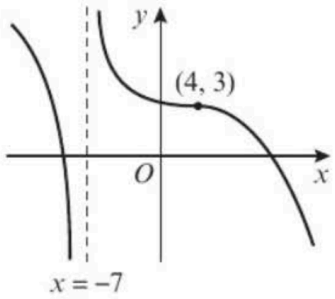
a



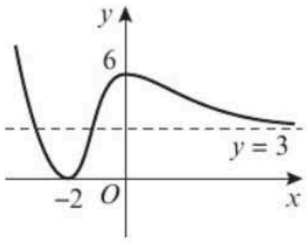
b



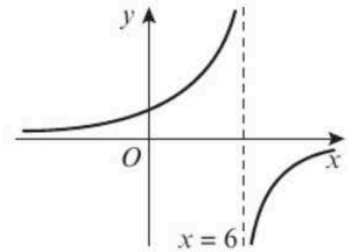
c



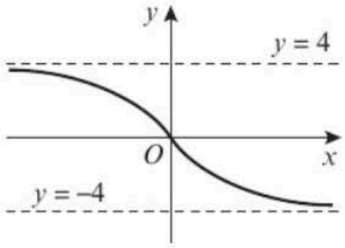
d



e



f



Worked solutions are available from this slide online.

2. $f(x) = (x + 1)(x - 4)^2$.
   a. Sketch the graph of $y = f(x)$.
   b. On a separate set of axes, sketch the graph of $y = f\prime(x)$.
   c. Show that $f\prime(x) = (x - 4)(3x - 2)$.
   d. Use the derivative to determine the exact coordinates where the gradient function cuts the coordinate axes.

a.

b.

c.

d.

Extension — Luisa · 69

[MAT 2015 1B] $f(x) = (x + a)^n$, where a is real and n is an integer with $n \geq 2$. How many intersections do $y = f(x)$ and $y = f\prime(x)$ have?

(a) Always an odd number. (b) Always an even number.

(c) Depends on a only. (d) Depends on n only. (e) Depends on both.

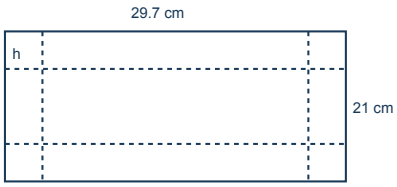
Answer

Factor the equation rather than dividing by a power of x + a, which would lose the intersection at x = −a.

Optimisation Problems/Modelling — Luisa · 70

Optimisation means choosing a variable to maximise or minimise a quantity.

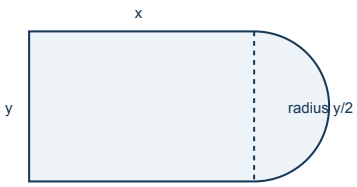
1. An open cuboid from A4 paper



Cut squares of side h from the four corners of a 21 cm × 29.7 cm sheet, then fold up the sides. Which h maximises the volume?

1

2. A pen with a semicircular end



A rectangular section x by y has a semicircle of diameter y at one end. The outer fence has length 50 m. Which x and y maximise the enclosed area?

2

‘Rate of change’ **Luisa · 71**

A derivative is a rate of change

dy/dx means the rate at which y changes with respect to x. The same notation works for other physical quantities.

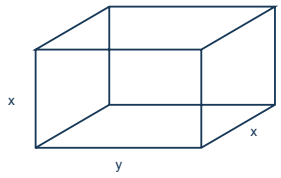
A sewage container fills at 20 cubic centimetres per second. Use appropriate notation to represent this.

Rate

Here V is volume and t is time. The derivative describes the rate at which the volume changes with time.

Example Optimisation Problem **Luisa · 72**

An open-topped cuboid tank is made from 54 m^2 of sheet metal. Its height is x metres and two opposite vertical faces are squares. Let the remaining dimension be y metres.



(a) Show that $V = 18x - \frac{2}{3}x^3$.

(b) Find and justify the maximum volume.

a

b

Test Your Understanding — Luisa · 73

A cuboid has a rectangular cross-section whose length is twice its width, x cm, as shown in Figure 2. The volume is 81 cm^3.

The total length of the twelve edges is L cm.

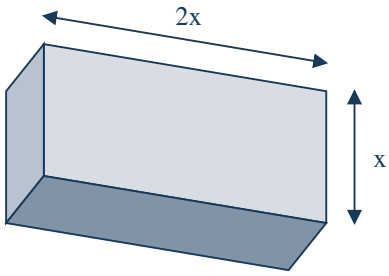


Figure 2

a. Show that $L = 12x + \frac{162}{x^2}$. (3 marks)

b. Use calculus to find the minimum value of L. (6 marks)

c. Justify, by further differentiation, that your value of L is a minimum. (2 marks)

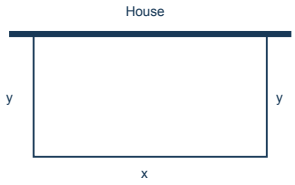
a.

b.

c.

Exercise 12K – Pearson Pure Y1 page 281 — Luisa · 74

6. A rectangular garden is fenced on three sides; the house forms the fourth side. The fence is 80 m long. Let y be the distance from the house and x the width.



(a) Show that A = y(80 − 2y).

(b) Find the dimensions and maximum area.

a

b

Exercise 12K – Pearson Pure Y1 page 281 **Luisa · 75**

7. A closed cylinder has total surface area 600π cm^2. Its radius is r cm and its height is h cm.

(a) Show that $V = 300\pi r - \pi r^3$.

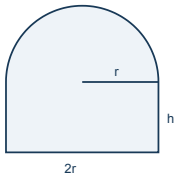
(b) Find the maximum volume.

a

b

Exercise 12K – Pearson Pure Y1 page 281 **Luisa · 76**

9. A shape consists of a rectangle with a semicircular top. Its perimeter is 40 cm; the semicircle has radius r cm.



(a) Show that $A = 40r - 2r^2 - \frac{1}{2}\pi r^2$. (2 marks)

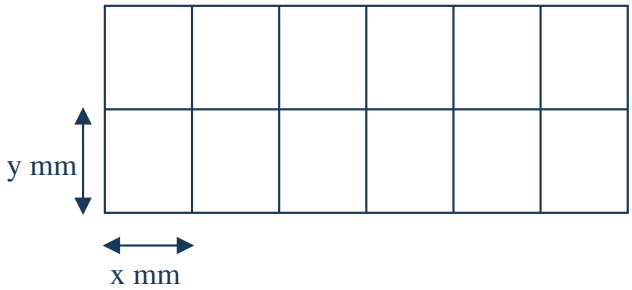
(b) Hence find the maximum area. (4 marks)

a

b

Exercise 12K – Pearson Pure Y1 page 281 **Luisa · 77**

10. A wire frame forms a large rectangle divided into 12 equal rectangles, in six columns and two rows. Each small rectangle has width x mm and height y mm. The total length of wire is 1512 mm.



a. Show that the total area is $A = 1296x - \frac{108x^2}{7}$ mm^2. (4 marks)

b. Find the maximum area enclosed by the frame. (4 marks)

a.

b.

Extension **Luisa · 78**

[STEP I 2006 Q2] A goat is tethered at ground level to a side of a square barn of side 2a in a large field. Its rope has length 4a. If A is the grazing area, prove that $A \leq 14\pi a^2$ and find its minimum.

Let x be the distance from one corner, with $0 \leq x \leq 2a$. The accessible area is a semicircle in front of the barn and four quarter-circle sectors reached around its corners.

Find the bounds